Google Professional Machine Learning Engineer Training - Activities

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Each activity is self-contained with a detailed printable guide, mock data or configuration, scripts, acceptance checks and cleanup guidance.

- Activity 01 - Create a Safe Google Cloud Training Environment - Create a Google account, Cloud project, billing budget, IAM boundary and Cloud Shell workspace.
- Activity 02 - Draft an ML Architecture Decision Record - Convert business volume, latency, security and explainability needs into a service decision.
- Activity 03 - Train and Evaluate a BigQuery ML Churn Model - Run SQL-defined logistic regression, evaluation and batch prediction on mock customer data.
- Activity 04 - Design an AutoML Tabular Experiment - Create a dataset specification, split plan, budget and evaluation slice gate.
- Activity 05 - Compare Pretrained Google Cloud AI APIs - Prepare Vision, Document AI and Natural Language requests and interpret mock responses.
- Activity 06 - Build a Data Quality and Feature Pipeline - Validate schema, repair missing values, encode features and preserve a training manifest.
- Activity 07 - Train a TensorFlow/Keras Model - Use tf.data, a Keras network, early stopping and SavedModel verification.
- Activity 08 - Package a Vertex AI Custom Training Job - Build a reproducible training package and container with managed output paths.
- Activity 09 - Analyze Vertex AI Hyperparameter Trials - Define a Vizier-style search space and select a trial using quality and cost evidence.
- Activity 10 - Plan Distributed PyTorch GPU Training - Configure torchrun-style workers, checkpoints and scaling-efficiency diagnostics.
- Activity 11 - Verify Feature Store Point-in-Time Parity - Model offline/online feature definitions and detect stale or leaked features.
- Activity 12 - Register and Canary a Model Version - Create registry metadata, traffic split, guardrails and rollback evidence.
- Activity 13 - Serve a Model on Cloud Run - Package a Flask prediction API, test its contract and analyze load-test results.
- Activity 14 - Compile a Vertex AI Pipeline - Connect validate, transform, train, evaluate and conditional deployment components.
- Activity 15 - Design an ML CI/CD/CT Release - Create Cloud Build gates, artifact provenance and a model card release record.
- Activity 16 - Investigate Drift, Bias and an ML Incident - Calculate drift and slice metrics, enforce SLOs and produce an incident evidence pack.